R22

27/11/2025

Code No.: 22ADE12

CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY (Autonomous)
B.E. (CSE) V Sem (Main/Backlog) Examination November 2025

Big Data Analytics

Time: 3 Hours **Max Marks: 60**

Note: "Answer **ALL** questions from **Part-A** and **Part-B** (including **internal choices**) at one place and in the same order as in the question paper."

Part - A
(5Q X 2M = 10 Marks)

			M	CO	BT
1		Differentiate between the Hadoopfile system and the traditional file system.	(2)	1	2
2		What are user-defined functions in Pig?	(2)	2	1
3		What is the difference between transformations and actions in Spark?	(2)	3	2
4		What is PageRank in GraphX?	(2)	4	2
5		Write the syntax for reading a streaming dataset from Kafka in Spark.	(2)	5	1

Part - B
(5Q X 10M = 50 Marks)

			M	CO	BT
6	(a)	Explain how Big Data differs from traditional data processing.	(5)	1	2
	(b)	Describe the architecture of the Hadoop stack with a neat diagram.	(5)	1	3
		(OR)			
7	(a)	Design MapReduce program for word count and explain.	(5)	1	4
	(b)	Design the steps in developing MapReduceapplication with an example.	(5)	1	4
8	(a)	Describe the architecture of Hive with a neat diagram.	(5)	2	3
	(b)	Explain how Pig can be used in practice for data processing?	(5)	2	2
		(OR)			
9	(a)	Explain how user-defined functions enhance Hive query capabilities with examples.	(3)	2	2
	(b)	Design a sample HiveQL script for querying student marks data.	(7)	2	4
10	(a)	Explain in detail how RDDs are created from different data sources with examples.	(5)	3	2
	(b)	Describe the process of creating and querying DataFrames in Spark SQL with examples.	(5)	3	3
		(OR)			
11	(a)	Compare and contrast operations and transformations in Spark with examples.	(5)	3	4
	(b)	State the benefits and challenges of using Spark for big data processing?	(5)	3	4
12	(a)	Describe the steps of designing a machine learning system in Spark.	(3)	4	2
	(b)	Build a classification model with spark.	(7)	4	2
		(OR)			
13	(a)	Explain how to obtain and preprocess data using Spark before model building.	(5)	4	3
	(b)	Explain with an example how regression model is built in Spark.	(5)	4	2

14	(a)	Explain the architecture of Spark structured streaming with a suitable diagram.	(5)	5	3
	(b)	Discuss a integration of Kafka with Spark, and Flink for advanced stream analytics.	(5)	5	4
		(OR)			
15	(a)	Explain how real-time monitoring can be implemented using Kafka with an example.	(7)	5	2
	(b)	Compare the features of batch processing and stream processing in terms of latency and throughput.	(3)	5	2

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